

MIS780 Advanced AI For Business - Assessment 1 - T2 2025

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Executive Summary

Aim

This project aims to support Insight IQ's airline client in understanding customer feedback through sentiment and topic analysis of 50,000 airline reviews. The objective is to extract actionable insights that highlight strengths, weaknesses, and emerging trends to inform service improvement and enhance customer satisfaction.

Business Problem

The client operates in a highly competitive airline market where customer experience significantly influences brand loyalty and revenue. While vast amounts of review data are collected, they remain underutilised for strategic decisions. The core business problem is the lack of structured insights from this unstructured text data to identify performance gaps, customer concerns, and improvement opportunities across service categories and cabin types.

Proposed Approaches

To address this, the project utilised natural language processing and data mining techniques:

- Sentiment Analysis using the VADER lexicon to quantify positive, negative, and neutral sentiment across airlines and cabin classes.
- Topic Modelling via Latent Dirichlet Allocation (LDA) to uncover major themes and concerns expressed by passengers.

Additional analyses included time-series comparisons, top airline rankings, and evaluations of entertainment, food, and ground services across cabins.

Major Findings

- Airlines ranking high overall (Section A) did not consistently perform well across all service aspects. For instance, discrepancies in food or ground services suggest targeted improvements could strengthen brand perception.
- Qantas Airways showed notable variability in ratings over time (Section B), especially around industry events such as the Boeing 737 MAX grounding and the COVID-19 pandemic.
- Sentiment analysis revealed that some airlines receive highly polarised feedback, while others maintain consistently positive impressions (Sections C & D). Qantas had strong sentiment in Business Class service, whereas some competitors outperformed in Economy.
- Topic modelling identified key concerns such as delays, comfort, staff behaviour, and pricing (Section E). Importantly, differences in concerns between top- and bottom-rated airlines revealed that lower-rated carriers faced more issues with staff and punctuality (Section F).

These findings provide a data-driven foundation for improving customer service, tailoring marketing strategies, and refining operational focus to better meet passenger expectations and drive competitive advantage.

```
In [1]: #connecting to drive
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [2]: #Loading libraries required
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: #importing dataset
excel_file_path = '/content/drive/MyDrive/Colab Notebooks/MIS780/A1_dataset.csv'
df = pd.read_csv(excel_file_path)

seed = 224142965
df_airline = df.sample(n=50000, random_state=seed)
num_rows, num_columns = df_airline.shape
print(f"Number of rows: {num_rows}")
print(f"Number of columns: {num_columns}")

df_airline.head()
```

Number of rows: 50000

Number of columns: 17

Out[3]:

	AirlineName	CabinType	DateFlown	EntertainmentRating	FoodRating	GroundServiceRating
59798	Spirit Airlines	Economy Class	Aug-15	0	1	2
877	Air Berlin	Economy Class	Oct-16	3	1	1
35083	Jet Airways	Economy Class	Dec-17	0	1	2
54591	Sun Country Airlines	Economy Class	Nov-19	0	0	1
64990	Turkish Airlines	Economy Class	Sep-22	0	3	1

In [4]:

```
# Check for missing values
missing = df_airline.isnull().sum()
missing = missing[missing > 0]

if not missing.empty:
    print("Missing values found in the following columns:")
    print(missing)
else:
    print("No missing values found.")

# Check for duplicate rows
print("Number of duplicate rows:", df_airline.duplicated().sum())
```

No missing values found.
Number of duplicate rows: 0

In [5]:

```
# Print the dataframe data types
print("\nDataFrame data types:")
print(df_airline.dtypes)

df_airline.describe()
```

```
DataFrame data types:
AirlineName      object
CabinType        object
DateFlown        object
EntertainmentRating  int64
FoodRating       int64
GroundServiceRating int64
OriginCountry    object
OverallScore     int64
Recommended      object
Review           object
SeatComfortRating int64
ServiceRating    int64
Title            object
TravelType       object
ValueRating      int64
WifiRating       int64
unique_id        int64
dtype: object
```

Out[5]:

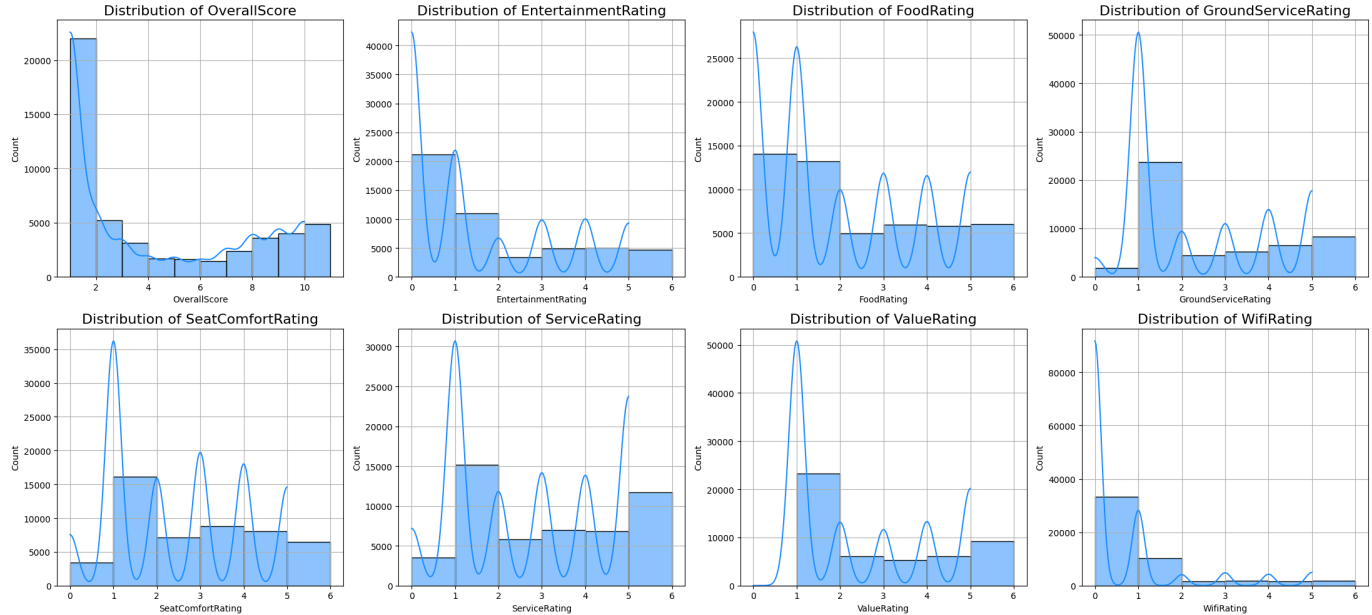
	EntertainmentRating	FoodRating	GroundServiceRating	OverallScore	SeatComfortRating	Sei
count	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	5
mean	1.512620	1.885720	2.316240	3.913920	2.430440	
std	1.733949	1.743124	1.628007	3.414824	1.548796	
min	0.000000	0.000000	0.000000	1.000000	0.000000	
25%	0.000000	0.000000	1.000000	1.000000	1.000000	
50%	1.000000	1.000000	1.000000	2.000000	2.000000	
75%	3.000000	3.000000	4.000000	7.000000	4.000000	
max	5.000000	5.000000	5.000000	10.000000	5.000000	

```
In [6]: # distributions of numerical variables
numerical_cols = [
    'OverallScore', 'EntertainmentRating', 'FoodRating', 'GroundServiceRating',
    'SeatComfortRating', 'ServiceRating', 'ValueRating', 'WifiRating'
]

fig, axes = plt.subplots(2, 4, figsize=(22, 10))
axes = axes.flatten()

for i, col in enumerate(numerical_cols):
    sns.histplot(df_airline[col], bins=range(df_airline[col].min(), df_airline[col].max()+2),
    axes[i].set_title(f'Distribution of {col}', fontsize=16)
    axes[i].set_xlabel(col)
    axes[i].set_ylabel('Count')
    axes[i].grid(True)

plt.tight_layout()
plt.show()
```



Data Exploration

EDA on numerical values

The figure above presents the distributions of key rating attributes in the airline review dataset, including Overall Score, Entertainment, Food, Ground Service, Seat Comfort, Service, Value, and WiFi.

Overall Score

The distribution of the overall rating is highly **right-skewed**, with a large concentration of reviews assigning low scores, primarily between **1 and 2**. Despite this, a noticeable segment of reviews provides high ratings in the **8 to 10** range.

This polarization suggests that passengers tend to have either very poor or very positive experiences, with fewer moderate responses. Such a trend may reflect **inconsistencies in service quality** across flights, routes, or airline providers.

Entertainment, Food, Ground Service, Seat Comfort, Service, Value, and WiFi

All subcategory attributes follow a similar pattern to the Overall Score, with distributions skewed toward the lower end. Most ratings cluster around **1 or 2 out of 5**, indicating **widespread dissatisfaction** with these aspects of the flight experience.

- **WiFi** stands out with the largest concentration of **zero and one** ratings, strongly suggesting that **poor or unavailable in-flight WiFi** is a frequent issue across airlines.
- **Service and Ground Service** exhibit a **bimodal distribution**, with a significant number of reviews falling at both the low and high ends of the scale. This pattern implies that service experiences **vary significantly**, potentially due to differences between airlines, crew members, or service classes.
- **Value for Money** is another aspect where dissatisfaction is pronounced. The majority of ratings are clustered at the **lowest point**, implying that most passengers do not feel they receive **good value** for the cost of their tickets. Only a small minority rate value positively.

- **Food and Entertainment** ratings, while somewhat more spread out, still lean heavily toward the **lower end**. This points to common dissatisfaction with the **quality or availability** of in-flight amenities.
- **Seat Comfort** follows the overall trend, with ratings most commonly at the low end. This underscores ongoing issues with **cabin ergonomics, space, or seating design**.

Summary of Patterns and Implications

Across the board, **passenger dissatisfaction** is evident — not only in the overall rating but consistently across individual service attributes.

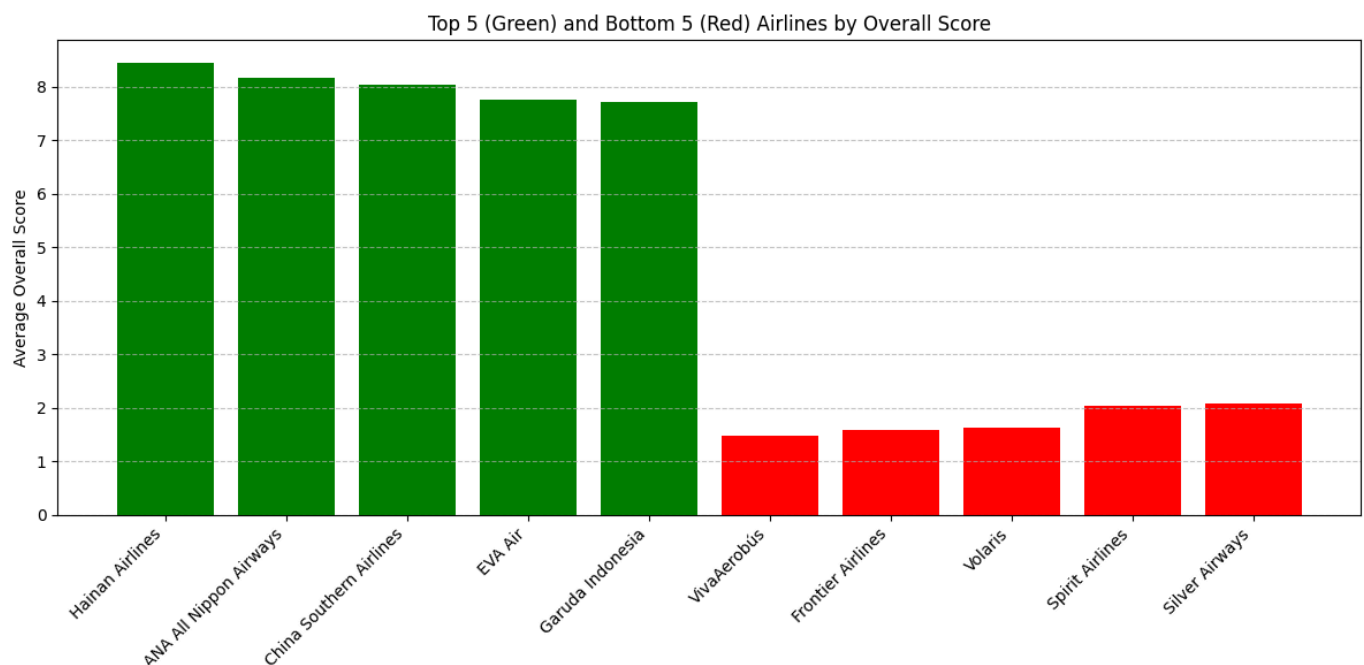
- The **recurrent right-skewed distributions** suggest that airlines may be falling short on **fundamental aspects** of the customer experience, particularly in **comfort, reliability, value, and in-flight amenities**.
- At the same time, the presence of **peaks in the high rating ranges** for some attributes indicates that certain airlines or flight services are delivering **exceptional experiences**. These positive reviews may correspond to **specific carriers, premium cabins, or routes with superior offerings**.

```
In [7]: # Calculate average OverallScore per airline
airline_scores = df_airline.groupby('AirlineName')['OverallScore'].mean()

# Get top 5 and bottom 5 airlines
top5_airlines = airline_scores.nlargest(5)
bottom5_airlines = airline_scores.nsmallest(5)

# Combine for plotting
combined = pd.concat([top5_airlines, bottom5_airlines])

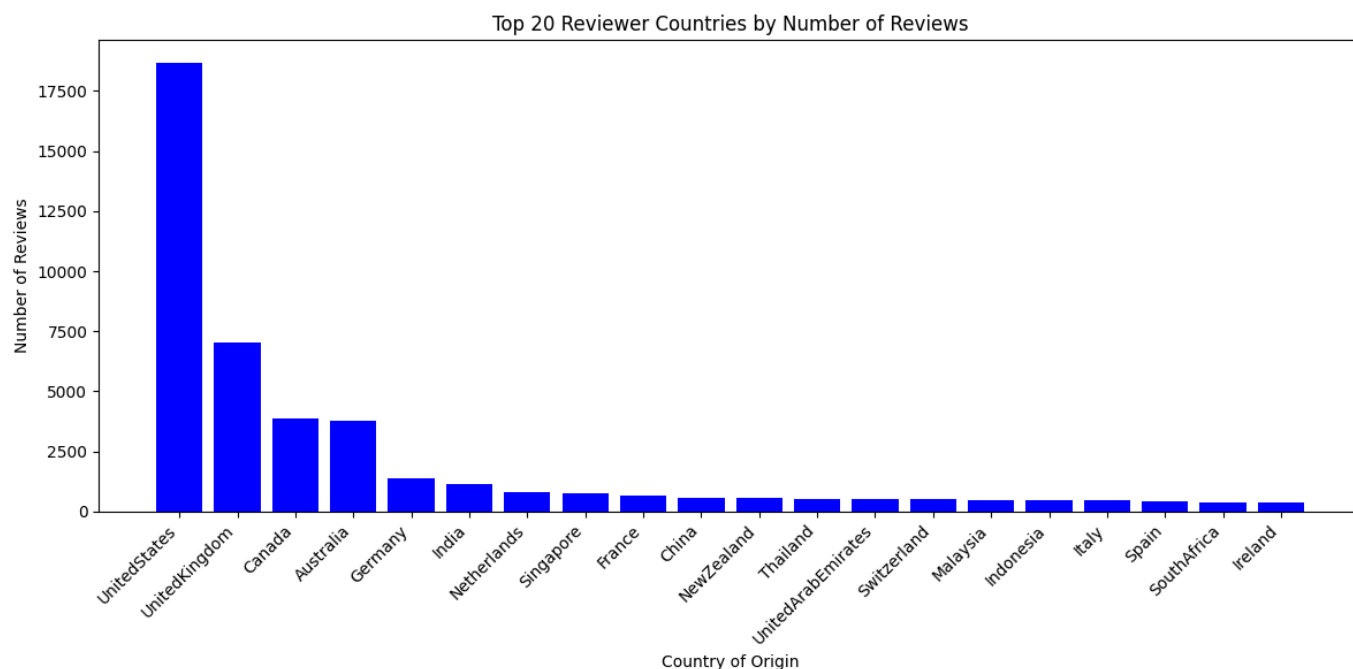
# Plot
plt.figure(figsize=(12,6))
bars = plt.bar(combined.index, combined.values, color=['green']*5 + ['red']*5)
plt.xticks(rotation=45, ha='right')
plt.ylabel('Average Overall Score')
plt.title('Top 5 (Green) and Bottom 5 (Red) Airlines by Overall Score')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



In [8]: *#Top 20 Reviewers' country of origin*

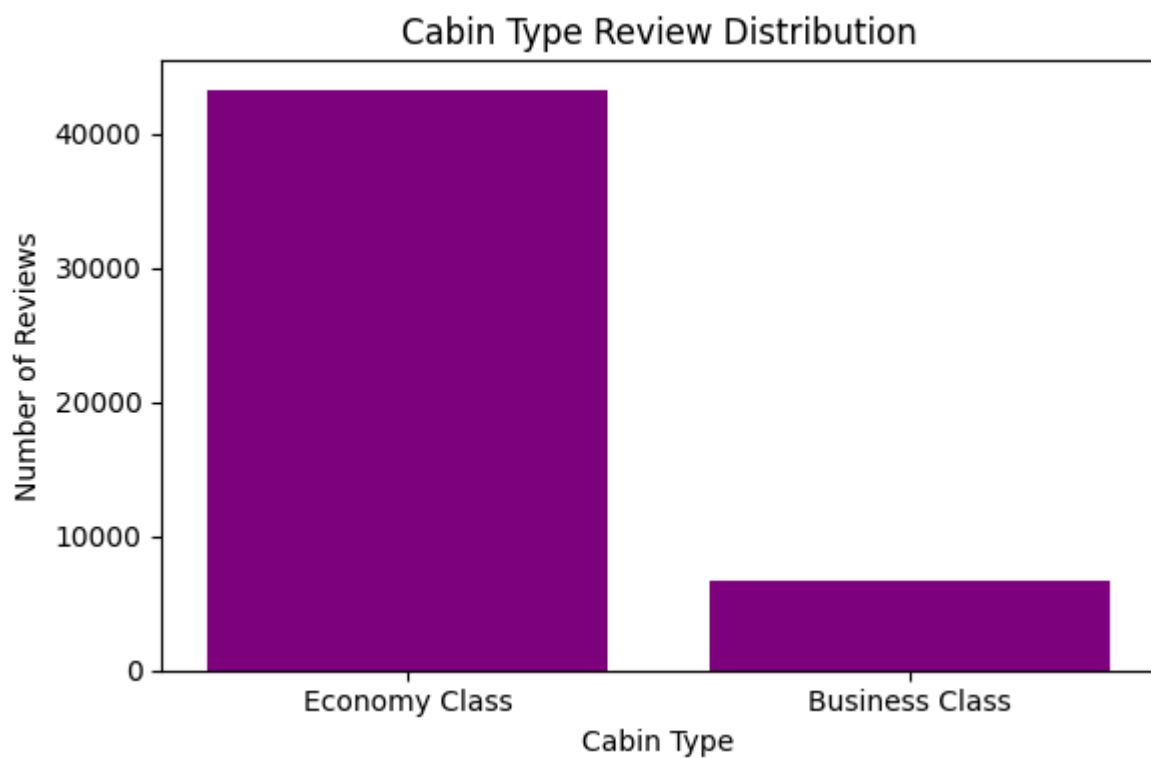
```
top_countries = df_airline['OriginCountry'].value_counts().head(20)

plt.figure(figsize=(12, 6))
plt.bar(top_countries.index, top_countries.values, color='blue')
plt.title('Top 20 Reviewer Countries by Number of Reviews')
plt.ylabel('Number of Reviews')
plt.xlabel('Country of Origin')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

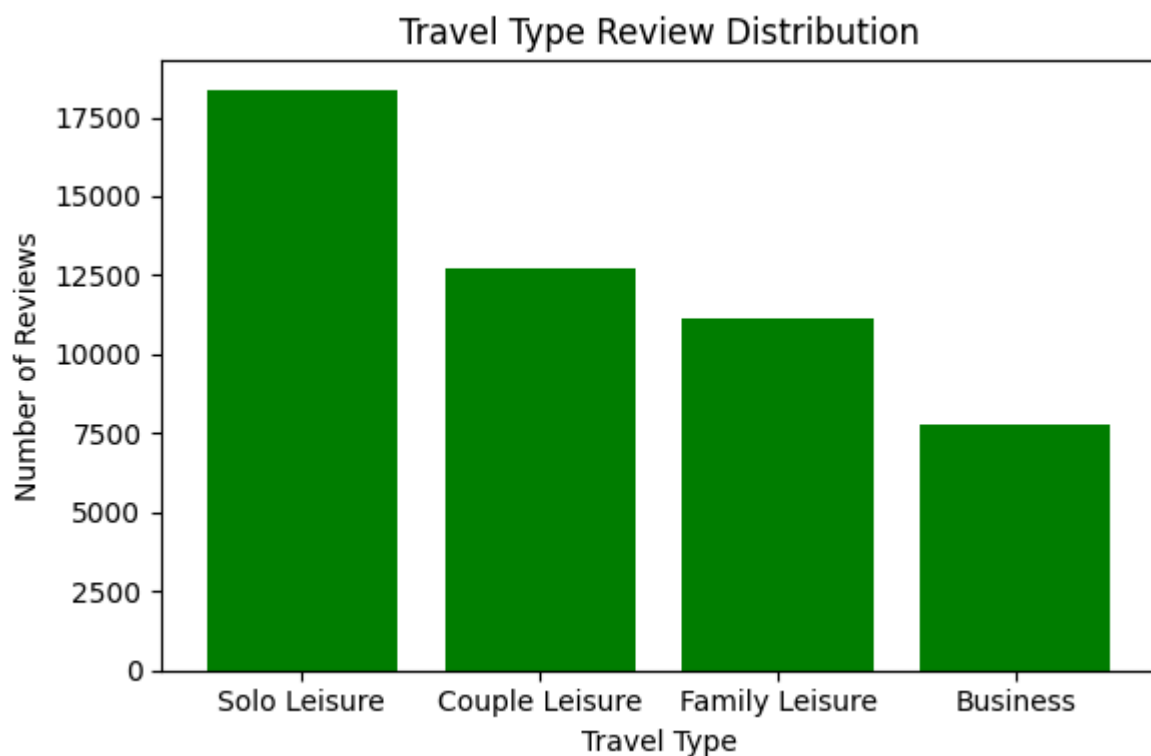


In [9]: *# Calculate review count by cabin type*

```
cabin_counts = df_airline['CabinType'].value_counts()
plt.figure(figsize=(6,4))
plt.bar(cabin_counts.index, cabin_counts.values, color='purple')
plt.title('Cabin Type Review Distribution')
plt.xlabel('Cabin Type')
plt.ylabel('Number of Reviews')
plt.tight_layout()
plt.show()
```



```
In [10]: # Calculate review count by travel type
travel_counts = df_airline['TravelType'].value_counts()
plt.figure(figsize=(6,4))
plt.bar(travel_counts.index, travel_counts.values, color='green')
plt.title('Travel Type Review Distribution')
plt.xlabel('Travel Type')
plt.ylabel('Number of Reviews')
plt.tight_layout()
plt.show()
```



Attribute Exploration & Insights

Top 5 vs Bottom 5 Airlines by Overall Score

The bar chart compares the average overall scores of the five highest-rated and five lowest-rated airlines.

- **Top-rated airlines:** Hainan Airlines, ANA All Nippon Airways, China Southern Airlines, EVA Air, Garuda Indonesia.
- **Lowest-rated airlines:** VivaAerobus, Frontier Airlines, Volaris, Spirit Airlines, Silver Airways.

The top performers received consistently high average ratings, indicating strong customer satisfaction and service quality. In contrast, the lowest-rated airlines scored significantly lower, highlighting persistent challenges in delivering acceptable service standards.

Insight:

The large performance gap between the best and worst airlines emphasizes the importance of consistency in service quality. Airlines with lower ratings should examine the practices of top performers and prioritize improvements in customer-facing services and operational reliability.

Top 20 Reviewer Countries by Number of Reviews

This chart shows which countries contribute the most reviews. The United States and the United Kingdom dominate in review volume, followed by Canada, Australia, and Germany.

Insight:

The prominence of English-speaking countries suggests cultural expectations and regional service standards may influence how passengers perceive and rate airline experiences. Airlines aiming to serve a global customer base should account for these differences in their customer service strategies.

Cabin Type Review Distribution

Economy Class accounts for the majority of reviews, while Business Class contributes a much smaller share.

Insight:

Since most customers travel in economy, improving the economy class experience can significantly impact overall satisfaction. However, given the premium nature of business class, even small improvements there may influence brand perception among high-value customers.

Travel Type Review Distribution

Solo Leisure is the most frequently mentioned travel type, followed by Couple Leisure and Family Leisure. Business travel is far less represented.

Insight:

Leisure travelers form the bulk of airline customers and should be the primary focus for service enhancements. Nonetheless, business travelers, though fewer in number, have higher expectations and may play a key role in shaping the airline's reputation.

Summary

The analysis reveals that the majority of airline passengers are economy-class leisure travelers, predominantly from a few countries. Service experiences vary significantly across airlines, with clear lessons to be learned from top-rated carriers. Improving mass-market services while maintaining high standards in premium offerings is key to boosting overall passenger satisfaction.

```

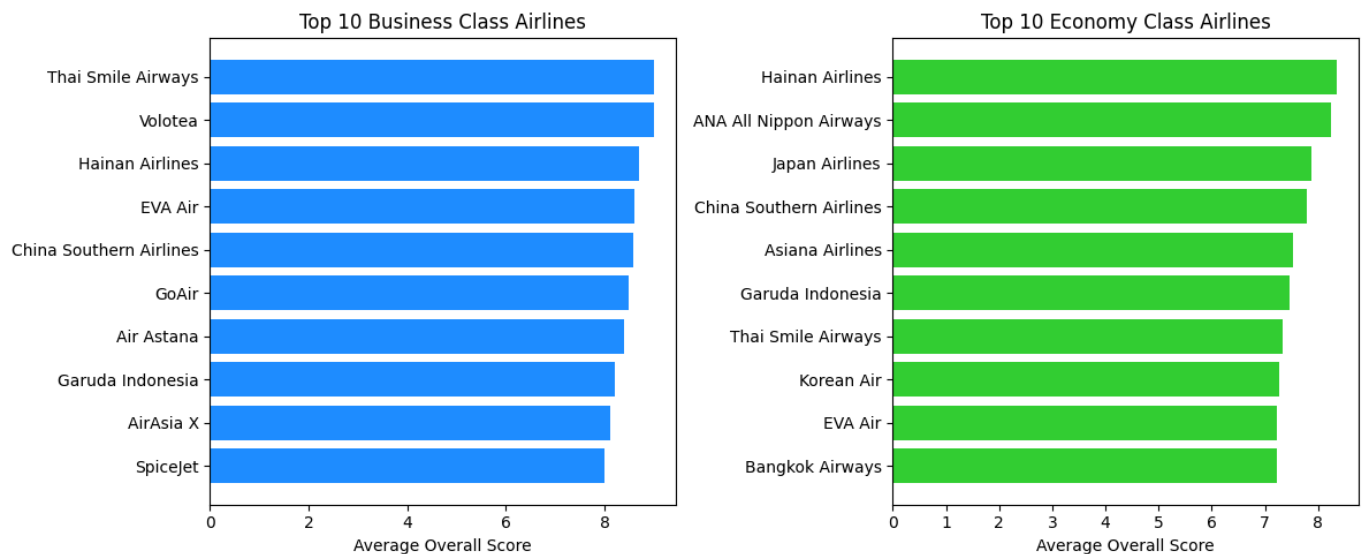
top10_business = df_airline[df_airline['CabinType']=='Business Class'].groupby('AirlineName')
top10_economy = df_airline[df_airline['CabinType']=='Economy Class'].groupby('AirlineName')

#bar plots
plt.figure(figsize=(12, 5))
plt.subplot(1,2,1)
plt.barh(top10_business.index[::-1], top10_business.values[::-1], color='dodgerblue')
plt.title('Top 10 Business Class Airlines')
plt.xlabel('Average Overall Score')

plt.subplot(1,2,2)
plt.barh(top10_economy.index[::-1], top10_economy.values[::-1], color='limegreen')
plt.title('Top 10 Economy Class Airlines')
plt.xlabel('Average Overall Score')

plt.tight_layout()
plt.show()

```



```

In [12]: # Top 10 airlines for each cabin type by Entertainment
top10_entertainment_business = df_airline[df_airline['CabinType']=='Business Class'].groupby('AirlineName')
top10_entertainment_economy = df_airline[df_airline['CabinType']=='Economy Class'].groupby('AirlineName')

# Top 10 airlines for each cabin type by Food
top10_food_business = df_airline[df_airline['CabinType']=='Business Class'].groupby('AirlineName')
top10_food_economy = df_airline[df_airline['CabinType']=='Economy Class'].groupby('AirlineName')

# Top 10 airlines for each cabin type by Ground Service
top10_ground_business = df_airline[df_airline['CabinType']=='Business Class'].groupby('AirlineName')
top10_ground_economy = df_airline[df_airline['CabinType']=='Economy Class'].groupby('AirlineName')

# Simple plots
plt.figure(figsize=(16, 10))

plt.subplot(2,3,1)
plt.barh(top10_entertainment_business.index[::-1], top10_entertainment_business.values[::-1], color='purple')
plt.title('Business - Entertainment')
plt.xlabel('Entertainment Rating')

plt.subplot(2,3,2)
plt.barh(top10_food_business.index[::-1], top10_food_business.values[::-1], color='red')
plt.title('Business - Food')
plt.xlabel('Food Rating')

plt.subplot(2,3,3)
plt.barh(top10_ground_business.index[::-1], top10_ground_business.values[::-1], color='purple')
plt.title('Business - Ground Service')
plt.xlabel('Ground Service Rating')

```

```

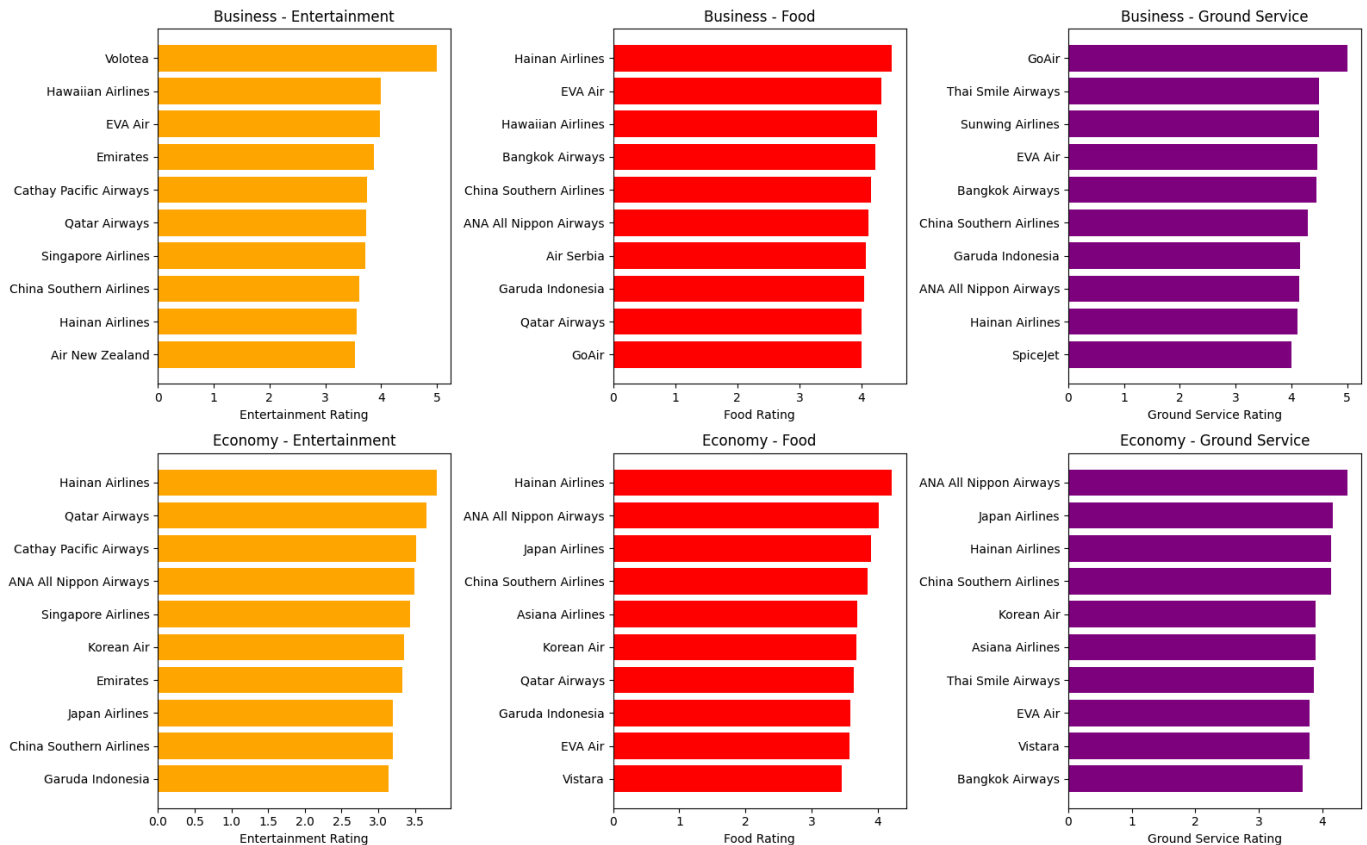
plt.subplot(2,3,4)
plt.barh(top10_entertainment_economy.index[::-1], top10_entertainment_economy.values[::-1], color='red')
plt.title('Economy - Entertainment')
plt.xlabel('Entertainment Rating')

plt.subplot(2,3,5)
plt.barh(top10_food_economy.index[::-1], top10_food_economy.values[::-1], color='red')
plt.title('Economy - Food')
plt.xlabel('Food Rating')

plt.subplot(2,3,6)
plt.barh(top10_ground_economy.index[::-1], top10_ground_economy.values[::-1], color='purple')
plt.title('Economy - Ground Service')
plt.xlabel('Ground Service Rating')

plt.tight_layout()
plt.show()

```



```

In [13]: # Convert 'Recommended' to numeric
df_airline['Recommended_flag'] = df_airline['Recommended'].map({'yes': 1, 'no': 0})

for cabin in df_airline['CabinType'].unique():
    recommended = df_airline[df_airline['CabinType'] == cabin].groupby('AirlineName')['Recommended'].head(10)
    top10 = recommended.head(10)
    bottom10 = recommended.tail(10)

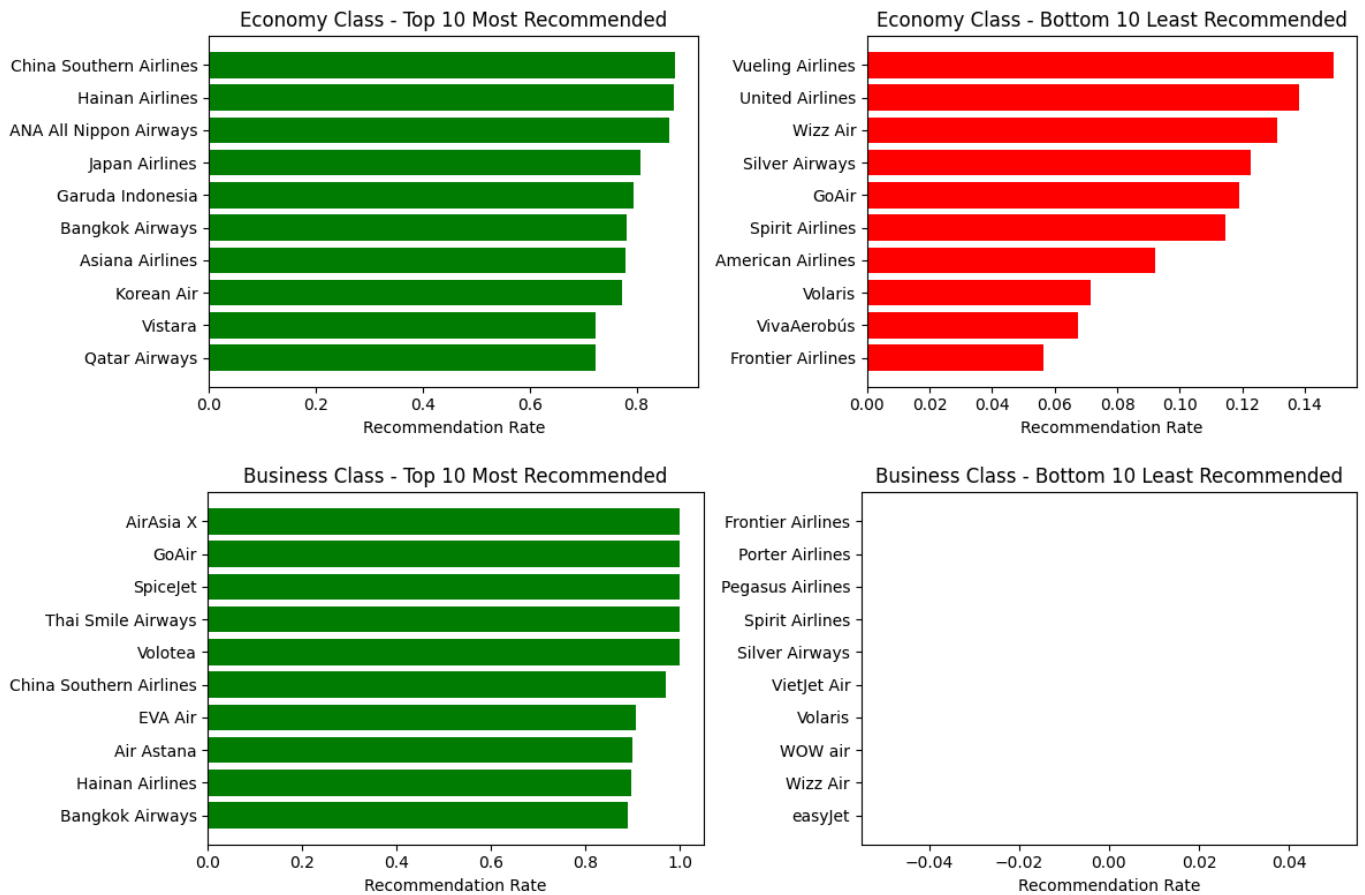
    plt.figure(figsize=(12,4))

    # Top 10
    plt.subplot(1,2,1)
    plt.barh(top10.index[::-1], top10.values[::-1], color='green')
    plt.title(f'{cabin} - Top 10 Most Recommended')
    plt.xlabel('Recommendation Rate')

    # Bottom 10
    plt.subplot(1,2,2)
    plt.barh(bottom10.index[::-1], bottom10.values[::-1], color='red')
    plt.title(f'{cabin} - Bottom 10 Least Recommended')
    plt.xlabel('Recommendation Rate')

```

```
plt.tight_layout()
plt.show()
```



Question A Analysis

Top 10 Airlines by Overall Rating

The analysis highlights clear differences between business and economy class leaders. In business class, **Thai Smile Airways** tops the list, followed by Volotea and Hainan Airlines. The full business class top 10 includes Thai Smile Airways, Volotea, Hainan Airlines, EVA Air, China Southern Airlines, GoAir, Air Astana, Garuda Indonesia, AirAsia X, and SpiceJet.

Economy class is led by **Hainan Airlines**, with ANA All Nippon Airways and Japan Airlines following. The economy top 10 are Hainan Airlines, ANA All Nippon Airways, Japan Airlines, China Southern Airlines, Asiana Airlines, Garuda Indonesia, Thai Smile Airways, Korean Air, EVA Air, and Bangkok Airways.

Performance Consistency Across Service Categories

Cross-Category Analysis

Some airlines show consistent strength across categories. **Hainan Airlines** leads economy overall and performs well in entertainment, food, and ground services in both cabins. China Southern Airlines, EVA Air, and Garuda Indonesia also appear frequently in top 10s, showing balanced service.

However, variations exist. For example, GoAir ranks first in business class ground service but is missing from entertainment and food top lists, suggesting strong airport operations but weaker in-flight amenities.

Service Category Breakdown

- **Entertainment:** Business class is led by Volotea, Hawaiian Airlines, and EVA Air; economy by Hainan Airlines, Qatar Airways, and Cathay Pacific. Entertainment leaders differ from overall rating leaders, indicating entertainment may weigh less in satisfaction scores.
- **Food:** Hainan Airlines and EVA Air top business class food; Hainan and ANA lead in economy. Food rankings align more closely with overall ratings, implying greater impact on passenger satisfaction.
- **Ground Services:** GoAir excels in business ground services, followed by Thai Smile Airways and Sunwing Airlines. Economy is led by ANA and Japan Airlines. Differences suggest ground service importance varies by passenger segment.

Regional and Operational Patterns

Asian airlines dominate most categories, especially carriers from China, Japan, South Korea, and Southeast Asia. This likely reflects better service models, cultural hospitality, or passenger expectations in key markets.

Performance differences may result from rating weightings, route-specific factors, airline investment focus, and varying passenger priorities.

Industry Structure Insight

Notably, the bottom 10 least recommended business class airlines have zero recommendation rates, implying they likely don't offer business class. Economy class carriers, even poorly rated, have some recommendations.

This suggests market segmentation: budget airlines focus on economy only, while full-service carriers compete across cabins. Airlines without business class tend to avoid that segment rather than offer subpar premium service.

Qantas Airways and Jetstar Airways Performance

Neither airline appears in any top 10 rankings (overall, entertainment, food, ground services) for either cabin class. They also don't feature in the bottom 10 least recommended, placing them in a middle performance tier.

Implications and Observations

Excellence demands balanced service across categories. Airlines top-rated overall perform consistently well in entertainment, food, and ground services. Those excelling in just one area fall short overall.

Qantas and Jetstar's middle-tier positioning shows no critical failures but highlights room for improvement. Asian carriers currently set the quality benchmark.

Sustainable competitive advantage requires systematic excellence across all customer touchpoints, not isolated investments in select services.

In [14]: *#question B*

```
# Convert DateFlown to datetime
df_airline['DateFlown'] = pd.to_datetime(df_airline['DateFlown'], format='%b-%y', errors='coe
```

```

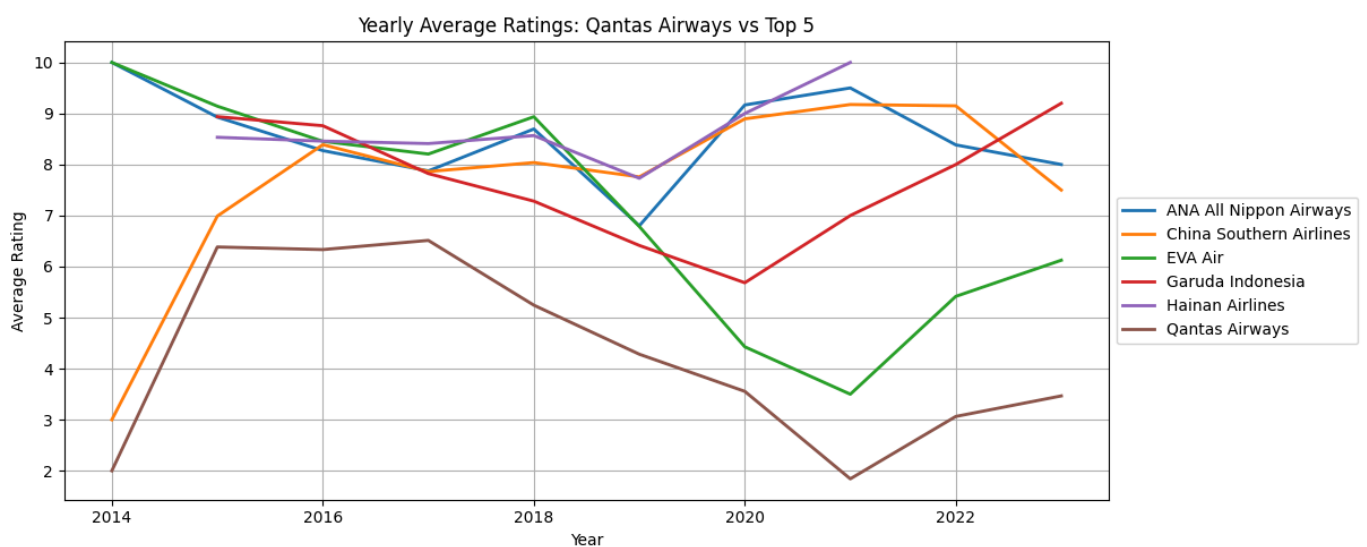
# Get top 5 airlines by average rating
top5 = df_airline.groupby('AirlineName')['OverallScore'].mean().nlargest(5).index.tolist()
selected = top5 + ['Qantas Airways']

# Filter for Qantas and top 5 only
df_selected = df_airline[df_airline['AirlineName'].isin(selected)].copy()
df_selected['Year'] = df_selected['DateFlown'].dt.year

# Calculate yearly average rating
result = df_selected.groupby(['AirlineName', 'Year'])['OverallScore'].mean().unstack(0)

# Plot
result.plot(figsize=(10,5), linewidth=2)
plt.title('Yearly Average Ratings: Qantas Airways vs Top 5')
plt.ylabel('Average Rating')
plt.xlabel('Year')
plt.grid(True)
plt.tight_layout()
plt.legend(loc='center left', bbox_to_anchor=(1, 0.5)) # <--- moves legend outside
plt.show()

```



Question B Analysis

Performance Trajectory Overview

Qantas Airways shows a distinctly different performance path compared to the top 5 airlines over 2014-2023, with key trends emerging.

Key Trends and Patterns

Qantas Airways Performance Evolution

2014-2017: Early Struggles and Recovery

- Started near 2.0 rating in 2014, well below competitors
- Improved steadily to about 6.5 by 2016
- Stabilized around 6.0-6.5 through 2017

2018-2021: Decline Phase

- Consistent drop post-2017
- Lowest ratings around 3.5-4.0 during 2019-2020

- Widening gap with top airlines

2021-2023: Modest Recovery

- Slight improvements since 2021
- Ratings remain low (~3.5-4.0) by 2023
- Still far behind leading airlines

Top 5 Airlines Comparative Performance

Consistent Leaders:

- **Hainan Airlines:** Stable ratings 8.5-10.0
- **ANA All Nippon Airways:** Strong 8.0-9.5 ratings
- **EVA Air:** Consistent 8.0-9.5 with low volatility

Fluctuating Performers:

- **China Southern Airlines:** Rose sharply 2014-2016, then fluctuated
- **Garuda Indonesia:** Experienced sharp dips around 2020 but recovered

Critical Divergence Points

2019-2020 Inflection:

- Top airlines stable or improving
- Qantas suffered steepest drop
- Largest performance gap occurred

2020-2021 COVID Impact:

- Hainan, EVA stable
- Garuda dipped then recovered
- Qantas saw prolonged negative impact

Potential Influencing Factors

Qantas-Specific Challenges

- Geographic isolation amplified pandemic effects
- Smaller domestic market, heavy reliance on international routes
- Fleet and workforce issues during border closures
- Service quality impacted by extended low utilization

Advantages of Top Performers

- Stronger regional/domestic markets for stability
- Cultural focus on service quality
- Consistent investment and adaptable operations
- Emphasis on premium service delivery

Industry-Wide Factors

- 737 MAX grounding and pandemic disruptions
- Supply chain issues
- Uneven global recovery and policy responses
- Intensified competition as markets reopened

Strategic Implications

Qantas faces structural challenges beyond temporary shocks. While top airlines maintained resilience and service quality, Qantas's prolonged decline points to deeper operational and strategic issues. Closing the gap requires sustained, broad improvements across services rather than relying on market recovery alone.

```
In [15]: #Question C
import nltk
nltk.download('vader_lexicon')
from nltk.sentiment.vader import SentimentIntensityAnalyzer
import matplotlib.pyplot as plt

sid = SentimentIntensityAnalyzer()

# Calculate average sentiment for each airline
scores = {}
for airline in df_airline['AirlineName'].unique():
    reviews = df_airline[df_airline['AirlineName'] == airline]['Review']
    avg = sum([sid.polarity_scores(str(r))['compound'] for r in reviews]) / len(reviews)
    scores[airline] = avg

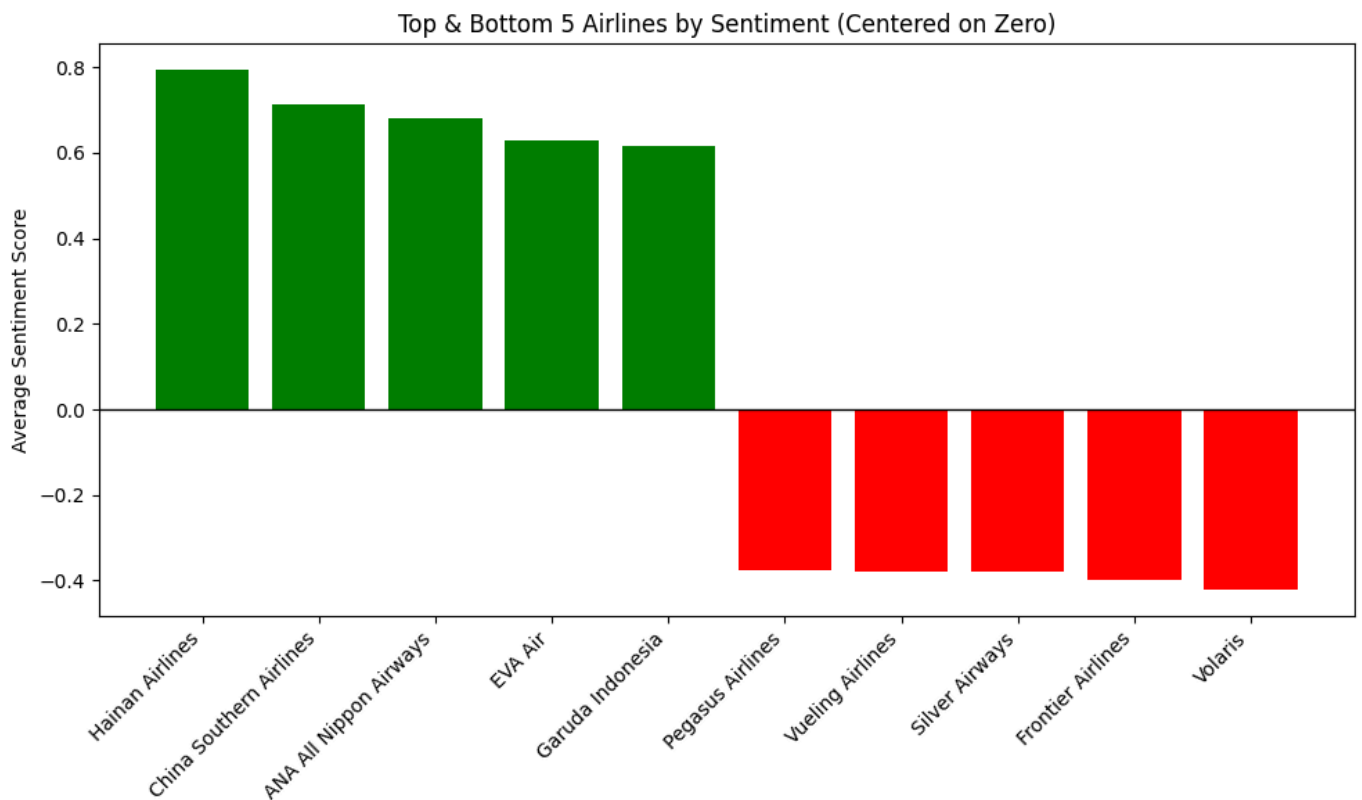
# Sort and get top 5 and bottom 5 airlines
sorted_scores = sorted(scores.items(), key=lambda x: x[1], reverse=True)
top5 = sorted_scores[:5]
bottom5 = sorted_scores[-5:]

# Combine into one list for plotting (top5 positive, bottom5 negative)
combined = top5 + bottom5
labels = [x[0] for x in combined]
values = [x[1] for x in combined]

# Color bars based on sign
colors = ['green' if v >= 0 else 'red' for v in values]

plt.figure(figsize=(10,6))
bars = plt.bar(labels, values, color=colors)
plt.axhline(0, color='black', linewidth=1)
plt.xticks(rotation=45, ha='right')
plt.ylabel('Average Sentiment Score')
plt.title('Top & Bottom 5 Airlines by Sentiment (Centered on Zero)')
plt.tight_layout()
plt.show()
```

[nltk_data] Downloading package vader_lexicon to /root/nltk_data...



Question C Analysis

Airline Sentiment Analysis

Most Positive Sentiment Performance

Hainan Airlines leads with the highest positive sentiment score (~0.8), topping the dataset.

Top 5 Airlines by Positive Sentiment

1. Hainan Airlines (~0.8)
2. China Southern Airlines (~0.72)
3. ANA All Nippon Airways (~0.68)
4. EVA Air (~0.62)
5. Garuda Indonesia (~0.61)

Sentiment Performance Spectrum

Positive Sentiment Leaders

Asian carriers dominate, with Chinese airlines occupying the top two spots, consistent with their strong service quality.

Negative Sentiment Airlines

Bottom 5 airlines show negative sentiment scores:

- Pegasus Airlines (~-0.35)
- Vueling Airlines (~-0.35)
- Silver Airways (~-0.35)

- Frontier Airlines (~-0.38)
- Volaris (~-0.42)

Key Observations

- **Sentiment Distribution:**
Top performers range 0.6-0.8; poor performers cluster around -0.35 to -0.42, showing clear polarization.
- **Regional Correlation:**
Asian full-service carriers dominate positives; low-cost European and American carriers feature in negatives.
- **Service Model Impact:**
Full-service airlines generally enjoy better sentiment than budget carriers.

Consistency with Service Quality Metrics

High sentiment rankings align with overall rating leaders (Hainan, China Southern, ANA). Poor service ratings coincide with negative sentiment, confirming the link between service quality and customer perception.

Strategic Implications

- Hainan Airlines exemplifies how consistent service excellence drives positive advocacy.
- Strong polarization suggests moderate performance yields neutral reactions, while airlines tend to evoke either positive or negative sentiments.
- Negative sentiment concentration among budget carriers highlights opportunities for differentiation via improved customer experience, even within cost constraints.

```
In [16]: # Create a simple subset with only AirlineName, CabinType, and Review columns
# Filter for reviews mentioning "service" or "services"

# Create a boolean mask for reviews containing "service" or "services"
service_mask = df_airline['Review'].str.lower().str.contains('service|services')

# Create the filtered DataFrame with columns required
df_service_reviews = df_airline.loc[service_mask, ['AirlineName', 'CabinType', 'Review']]

# Replace commas with periods in the Review column
df_service_reviews['Review'] = df_service_reviews['Review'].str.replace(',', '.')
```

```
In [17]: # Calculate sentiment scores for service reviews
df_service_reviews['sentiment'] = df_service_reviews['Review'].apply(lambda x: sid.polarity_scores(x)['polarity'])

# Group by 'AirlineName' and 'CabinType' and calculate average sentiment score
service_sentiment = df_service_reviews.groupby(['AirlineName', 'CabinType'])['sentiment'].mean()

# Find the airline with the highest service sentiment for Economy class
best_economy = service_sentiment[service_sentiment['CabinType'] == 'Economy Class'].sort_values(ascending=False)

# Find the airline with the highest service sentiment for Business class
best_business = service_sentiment[service_sentiment['CabinType'] == 'Business Class'].sort_values(ascending=False)

# Display results
print("\n----- AIRLINE WITH BEST SERVICE SENTIMENT -----")
if not best_economy.empty:
```

```

print(f"Economy Class: {best_economy.iloc[0]['AirlineName']} (sentiment score: {best_econ
else:
    print("No Economy Class reviews found.")

if not best_business.empty:
    print(f"Business Class: {best_business.iloc[0]['AirlineName']} (sentiment score: {best_bu
else:
    print("No Business Class reviews found.")

# Create simple bar chart
plt.figure(figsize=(10, 6))
labels = []
values = []
colors = []

if not best_economy.empty:
    labels.append('Economy Class - ' + best_economy.iloc[0]['AirlineName'])
    values.append(best_economy.iloc[0]['sentiment'])
    colors.append('blue')

if not best_business.empty:
    labels.append('Business Class - ' + best_business.iloc[0]['AirlineName'])
    values.append(best_business.iloc[0]['sentiment'])
    colors.append('red')

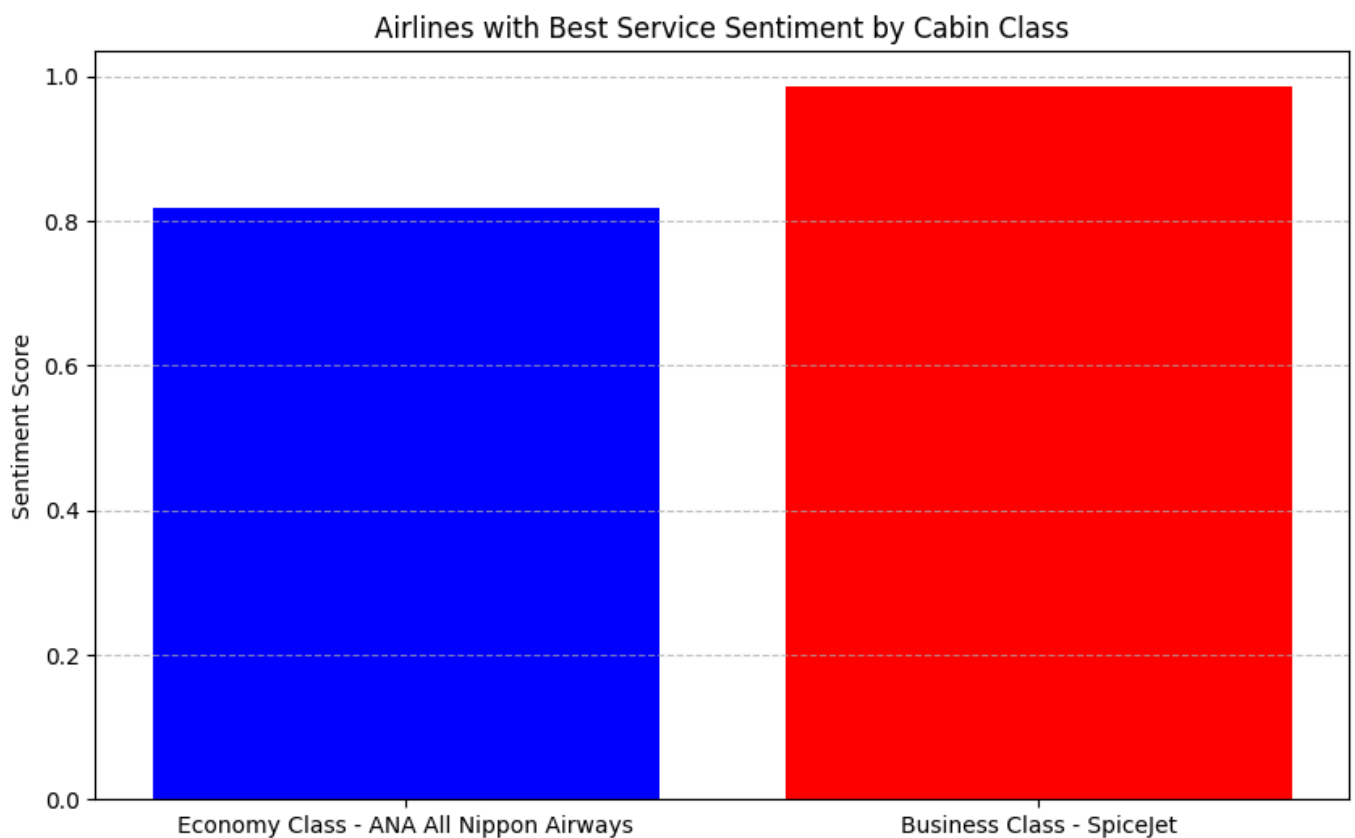
if labels: # Only plot if we have data
    plt.bar(labels, values, color=colors)
    plt.title('Airlines with Best Service Sentiment by Cabin Class')
    plt.ylabel('Sentiment Score')
    plt.grid(axis='y', linestyle='--', alpha=0.7)
    plt.show()

```

----- AIRLINE WITH BEST SERVICE SENTIMENT -----

Economy Class: ANA All Nippon Airways (sentiment score: 0.819)

Business Class: SpiceJet (sentiment score: 0.985)



Question D Analysis

Airlines with Best Service Sentiment by Cabin Class

Top Sentiment Scores by Cabin

- **Economy Class:** ANA All Nippon Airways — sentiment score ~0.82
- **Business Class:** SpiceJet — sentiment score ~0.98 (near-perfect)

Key Observations

- **Economy Class Excellence:**
ANA's high sentiment aligns with its strong overall and service category performance, reinforcing its reputation for quality in economy offerings.
- **Business Class Surprise:**
SpiceJet's near-perfect business class sentiment is exceptional, suggesting a highly differentiated premium product that exceeds expectations.
- **Sentiment Gap Between Cabins:**
SpiceJet's 0.98 score significantly surpasses ANA's 0.82, indicating that peak satisfaction can vary sharply between cabin types and carriers.

Implication

This analysis shows that excellence in one cabin class doesn't guarantee equivalent performance in another. Different airlines may lead in distinct segments, highlighting the importance of segment-specific service strategies.

```
In [21]: from nltk.stem import PorterStemmer
import re
import nltk
# Download the required NLTK data
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger_eng')
from nltk.corpus import stopwords
from nltk import word_tokenize, pos_tag
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.decomposition import LatentDirichletAllocation as LDA

# Initialize stemmer and download NLTK data
stemmer = PorterStemmer()

# 1. Cleaning raw reviews
raw_reviews = df_airline['Review']
cleaned_reviews = []

for i in range(len(raw_reviews)):
    text = raw_reviews.iloc[i]
    try:
        text = re.sub('[^A-Za-z]', ' ', text)
        text = text.lower()
        tokens = text.split()
        filtered_words = [word for word in tokens if len(word) > 3]
        cleaned_text = ' '.join(filtered_words)
    except:
        continue
```

```

cleaned_reviews.append(cleaned_text)

# 2. Remove stopwords
stop_words = stopwords.words('english')
for i in range(len(cleaned_reviews)):
    filtered_tokens = [word for word in cleaned_reviews[i].split() if word not in stop_words]
    cleaned_reviews[i] = ' '.join(filtered_tokens)

# 3. Keep only nouns using POS tagging
noun_only_reviews = []
for review in cleaned_reviews:
    if not isinstance(review, str):
        noun_only_reviews.append('')
        continue
    tokens = word_tokenize(review)
    tagged = pos_tag(tokens)
    nouns = [word for word, pos in tagged if pos.startswith('NN')]
    noun_only_reviews.append(' '.join(nouns))

# 4. Vectorize nouns with frequency filtering
vectorizer = CountVectorizer(max_df=0.2, min_df=0.01)
ReducedCount = vectorizer.fit_transform(noun_only_reviews)

# 5. Apply LDA
number_topics = 10
lda = LDA(n_components=number_topics, n_jobs=-1, random_state=2023)
lda.fit(ReducedCount)

```

```

[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.

```

Out[21]:

```

▼ LatentDirichletAllocation ⓘ ?
LatentDirichletAllocation(n_jobs=-1, random_state=2023)

```

In [26]:

```

from wordcloud import WordCloud
import matplotlib.pyplot as plt
import math

# Extract topic-word probabilities and terms from the fitted LDA model
Word_Topics_Pro = lda.components_
ReducedTerm = vectorizer.get_feature_names_out()

# Function to plot word clouds for topics
def plot_wordclouds(topics, terms, title):
    num_topics = len(topics)
    rows = math.ceil(num_topics / 4)

    fig, ax = plt.subplots(rows, 4, figsize=(15, 2.5 * rows))
    ax = ax.flatten() # flatten to 1D array for easy indexing

    # Turn off all axes initially
    for axi in ax:
        axi.axis('off')

    for topic_idx, topic in enumerate(topics):
        # Zip words and their probabilities, sort descending

```

```
count_dict = dict(sorted(zip(terms, topic), key=lambda x: x[1], reverse=True)[:10])

# Generate word cloud for top 10 words in the topic
wordcloud = WordCloud(background_color="white").generate_from_frequencies(count_dict)

# Plot on the respective subplot
ax[topic_idx].imshow(wordcloud, interpolation='bilinear')
ax[topic_idx].set_title(f"Topic {topic_idx + 1}")
ax[topic_idx].axis('off') # hide axis ticks

plt.suptitle(title, fontsize=16)
plt.tight_layout(rect=[0, 0, 1, 0.95]) # Adjust layout to prevent supitle overlap
plt.show()

# Plot word clouds for the topics
plot_wordclouds(Word_Topics_Pro, ReducedTerm, "Topics from Airline Reviews")
```

Topics from Airline Reviews



Question E Analysis

Key Passenger Concerns and Interests

Service Quality and Comfort

Passengers are deeply concerned about their in-flight experience, particularly cabin crew service, seating comfort, and amenities. Business class passengers have heightened expectations around service quality, while other travelers focus on meal quality, drinks, and entertainment systems. Crew performance emerges as a major factor in passenger satisfaction across all service levels.

Financial Concerns

Money matters significantly to passengers, with focus on ticket pricing, refunds, and booking processes. This extends to baggage-related costs and policies, suggesting passengers are increasingly frustrated with additional fees and complex baggage rules that impact their overall travel budget.

Luggage and Baggage Handling

Baggage appears as a major pain point across multiple areas of concern. Passengers worry about luggage handling, carry-on restrictions, additional charges, and the overall baggage experience. This

suggests airlines need to address both the practical and financial aspects of baggage services.

Operational Reliability

Passengers show significant concern about flight delays, connections, and schedule reliability. Hotel accommodations during delays become important, while family travelers have specific experiences with major carriers. There's notable anxiety about tight connections and boarding processes, particularly regarding gate changes and timing.

Destination-Specific Experiences

The prominence of Dubai, Sydney, Emirates, and Qantas suggests passengers have distinct expectations for long-haul international routes, with particular attention to transit experiences and lounge access at major hub airports.

Interesting Insights

Segmented Expectations

Business class passengers have fundamentally different concerns than economy passengers, focusing more on premium service elements.

Technology Integration

The appearance of "phone" and "information" suggests passengers want better digital communication about delays, gate changes, and service updates.

Family Travel

Specific mention of family-related terms indicates parents have unique concerns about traveling with children.

Hub Airport Experiences

Passengers pay attention to specific airport and airline combinations, suggesting route and hub quality significantly impact overall satisfaction.

Conclusion

These topics reveal that passenger concerns extend far beyond just getting from point A to point B - they encompass the entire travel ecosystem, from booking to baggage claim, with particular sensitivity to service quality, costs, and reliability. Airlines should focus on these core areas to improve passenger satisfaction and loyalty.

```
In [25]: from wordcloud import WordCloud
import math

# 1. Identify top 3 and bottom 3 airlines by average OverallScore
airline_scores = df_airline.groupby('AirlineName')['OverallScore'].mean()
top3_airlines = airline_scores.nlargest(3).index.tolist()
bottom3_airlines = airline_scores.nsmallest(3).index.tolist()

print("Top 3 Airlines:", top3_airlines)
```

```

print("Bottom 3 Airlines:", bottom3_airlines)

# 2. Filter noun_only_reviews for top 3 airlines and bottom 3 airlines
# Map cleaned reviews back to airlines
reviews_airlines = df_airline['AirlineName'].reset_index(drop=True)
# Make sure noun_only_reviews aligns index-wise with df_airline
assert len(noun_only_reviews) == len(df_airline)

# Filter reviews for top3 and bottom3 airlines
top3_reviews = [noun_only_reviews[i] for i in range(len(noun_only_reviews)) if reviews_airline[i] in top3_airlines]
bottom3_reviews = [noun_only_reviews[i] for i in range(len(noun_only_reviews)) if reviews_airline[i] in bottom3_airlines]

# 3. Define a function to vectorize and fit LDA
def lda_on_reviews(reviews, num_topics=5, max_df=0.2, min_df=0.01):
    vectorizer = CountVectorizer(max_df=max_df, min_df=min_df)
    dtm = vectorizer.fit_transform(reviews)
    lda_model = LDA(n_components=num_topics, random_state=2023)
    lda_model.fit(dtm)
    terms = vectorizer.get_feature_names_out()
    return lda_model, terms

# 4. Run LDA for top3 and bottom3 airlines
num_topics = 5

lda_top3, terms_top3 = lda_on_reviews(top3_reviews, num_topics)
lda_bottom3, terms_bottom3 = lda_on_reviews(bottom3_reviews, num_topics)

# 5. function to plot word clouds of topics
def plot_lda_wordclouds(lda_model, terms, title):
    topics = lda_model.components_
    num_topics = topics.shape[0]
    rows = math.ceil(num_topics / 4)
    fig, axes = plt.subplots(rows, 4, figsize=(15, 2.5 * rows))
    axes = axes.flatten()
    for ax in axes:
        ax.axis('off')

    for topic_idx, topic in enumerate(topics):
        word_probs = dict(sorted(zip(terms, topic), key=lambda x: x[1], reverse=True)[:10]))
        wc = WordCloud(background_color="white").generate_from_frequencies(word_probs)
        axes[topic_idx].imshow(wc, interpolation='bilinear')
        axes[topic_idx].set_title(f"Topic {topic_idx + 1}")
        axes[topic_idx].axis('off')

    plt.suptitle(title, fontsize=18)
    plt.tight_layout(rect=[0, 0, 1, 0.95])
    plt.show()

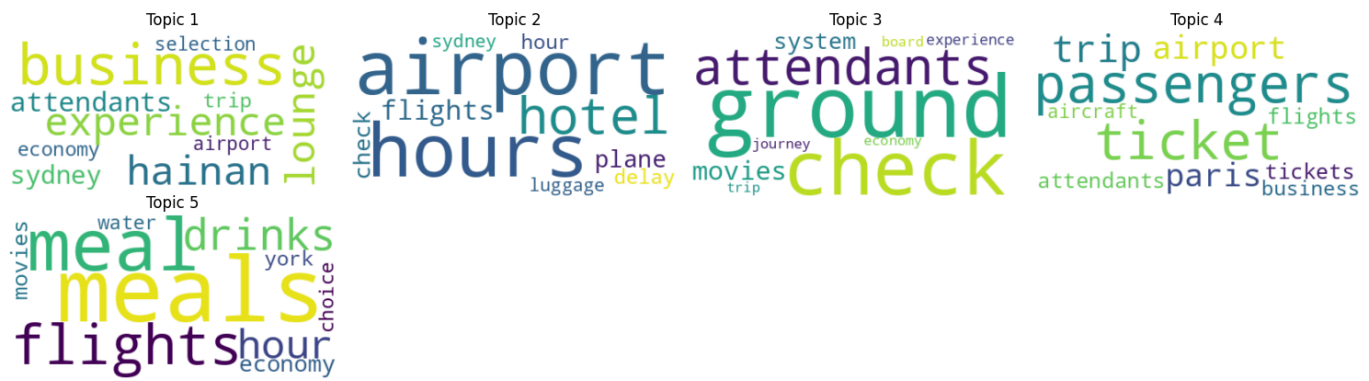
# 6. Visualize
plot_lda_wordclouds(lda_top3, terms_top3, "Top 3 Airlines Passenger Concerns and Interests")
plot_lda_wordclouds(lda_bottom3, terms_bottom3, "Bottom 3 Airlines Passenger Concerns and Interests")

```

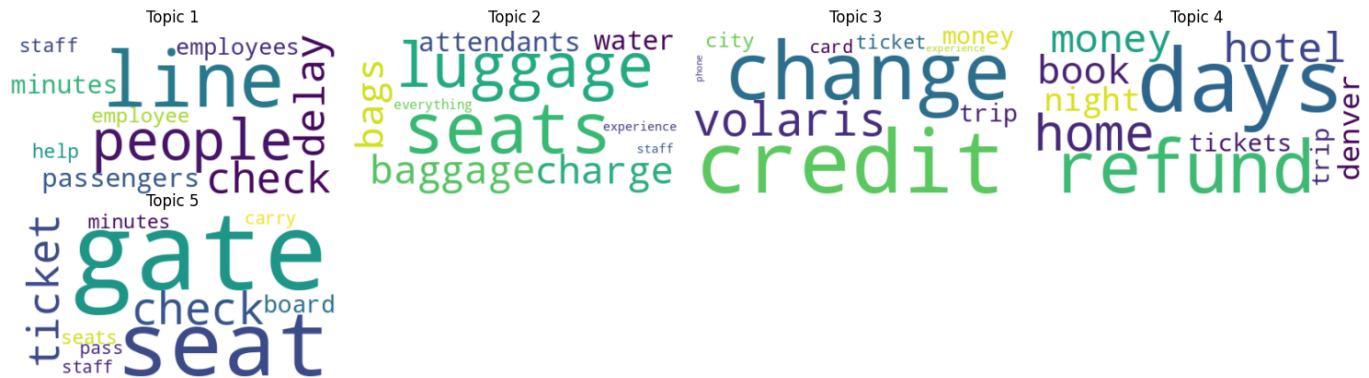
Top 3 Airlines: ['Hainan Airlines', 'ANA All Nippon Airways', 'China Southern Airlines']

Bottom 3 Airlines: ['VivaAerobús', 'Frontier Airlines', 'Volaris']

Top 3 Airlines Passenger Concerns and Interests



Bottom 3 Airlines Passenger Concerns and Interests



Question F Analysis

Differences in Passenger Concerns: Top 3 vs Bottom 3 Rated Airlines

The LDA topic modeling reveals striking differences in what passengers discuss when reviewing top-rated versus bottom-rated airlines. These differences highlight fundamental distinctions in service quality and passenger experience.

Key Differences in Passenger Concerns

Top 3 Airlines: Service Excellence Focus

Premium Service Experience

- Passengers discuss business class experiences, lounge access, and premium amenities
- Focus on positive service elements like meal quality, drinks, and in-flight entertainment
- Mentions of specific positive experiences with flight attendants and ground staff
- Discussion of airport facilities and smooth operations

Operational Efficiency

- References to specific routes (Sydney, Paris) suggesting successful long-haul operations
- Time-related discussions focus on flight duration rather than delays
- Airport and ground operations appear seamless in passenger narratives

Bottom 3 Airlines: Problem-Focused Discussions

Financial Disputes and Service Failures

- Heavy emphasis on refunds, credits, and money-related issues
- Frequent mentions of booking problems and ticket complications
- Significant focus on additional charges and unexpected fees

Operational Problems

- Extensive discussion of delays, gate changes, and scheduling issues
- Hotel accommodations needed due to service disruptions
- Staff-related complaints and service recovery failures
- Seating and boarding process difficulties

Topic-by-Topic Comparison

Business vs Basic Service

Top Airlines: Business class features prominently, indicating passengers experience and discuss premium services **Bottom Airlines:** Focus shifts to basic service failures and staff issues

Proactive vs Reactive Discussions

Top Airlines: Airport operations, flight planning, and journey management **Bottom Airlines:** Baggage problems, seat issues, and reactive problem-solving

Financial Satisfaction vs Financial Frustration

Top Airlines: Limited financial complaints, focus on value received **Bottom Airlines:** Dominant themes around refunds, credits, and financial disputes

Operational Smoothness vs Operational Chaos

Top Airlines: Time discussions relate to journey planning and efficiency **Bottom Airlines:** Time discussions relate to delays, extended waits, and disruptions

Potential Reasons for These Differences

Service Quality Investment

Top Airlines invest heavily in:

- Staff training resulting in positive attendant interactions
- Premium amenities that passengers actively discuss and appreciate
- Efficient operations that minimize complaints
- Proactive customer service that prevents issues

Bottom Airlines likely struggle with:

- Inconsistent staff performance leading to negative experiences
- Basic service failures that dominate passenger attention
- Reactive rather than proactive customer service
- Cost-cutting measures that impact service quality

Operational Excellence

Top Airlines demonstrate:

- Superior scheduling and on-time performance
- Effective baggage handling systems
- Streamlined boarding and gate operations
- Investment in ground services and facilities

Bottom Airlines show patterns of:

- Frequent delays and schedule disruptions
- Baggage handling problems
- Gate and boarding inefficiencies
- Poor coordination between departments

Customer Communication

Top Airlines appear to:

- Set and meet passenger expectations effectively
- Provide clear information about services and policies
- Handle transactions smoothly with minimal disputes

Bottom Airlines struggle with:

- Poor communication leading to booking confusion
- Unexpected charges and policy surprises
- Inadequate problem resolution processes
- Financial transaction difficulties

Financial Model Impact

Top Airlines likely:

- Price services appropriately, reducing financial complaints
- Provide value that justifies costs
- Handle refunds and credits efficiently

Bottom Airlines may:

- Rely on additional fees that frustrate passengers
- Create billing confusion and disputes
- Have inefficient refund and credit processes

Strategic Implications

The stark differences suggest that top-rated airlines succeed by:

1. **Investing in proactive service excellence** rather than reactive problem-solving
2. **Creating seamless operational experiences** that passengers notice and appreciate
3. **Maintaining transparent financial practices** that build trust
4. **Training staff effectively** to handle both routine and exceptional situations

Bottom-rated airlines appear trapped in cycles where service failures consume passenger attention and create negative word-of-mouth, while top airlines generate positive discussions about premium experiences and operational excellence.

This analysis suggests that airline success requires fundamental operational competence before premium service differentiation becomes relevant to passenger satisfaction.

5. Practical Implication

Recommendations for Supporting Passengers

Based on extensive analysis of passenger sentiment, service quality, and operational trends, the following recommendations can help airlines enhance passenger satisfaction and loyalty:

Immediate Priority Actions

1. Service Quality Fundamentals

To improve passenger satisfaction, airlines should focus on critical service aspects that heavily influence passenger *experiences*. These include upgrading economy class seating, as most feedback comes from this segment, implementing standardized and high-quality crew training modeled after top Asian carriers, investing in better WiFi infrastructure to address consistent dissatisfaction, and establishing service consistency frameworks to reduce the variability in passenger experiences.

2. Operational Reliability Enhancement

Operational disruptions remain a major source of passenger complaints, especially among lower-rated airlines. Airlines should use AI-driven predictive systems to anticipate delays and communicate proactively with passengers. End-to-end baggage tracking with proactive notifications will help reduce baggage issues. Transparent and real-time communication protocols during disruptions are essential to keep passengers informed and reduce anxiety. Additionally, streamlining ground services such as check-in, security, and boarding will reduce wait times and operational bottlenecks.

Strategic Competitive Positioning

3. Value Perception Transformation

Passenger dissatisfaction with perceived value for money suggests the need for clearer pricing and communication. Airlines should adopt transparent pricing models with all-inclusive base fares and clearly defined optional add-ons. Educating passengers about service investments and operational costs can improve understanding and acceptance. Clear differentiation between service tiers helps align expectations, while enhancing loyalty programs with tangible benefits rewards frequent travelers and strengthens brand loyalty.

4. Market-Specific Adaptation

Adapting service delivery to regional cultural expectations and market maturity enhances passenger satisfaction. This involves aligning service standards with cultural hospitality norms, providing market-specific training to crew members to improve communication and service, focusing investments on key routes with high passenger volume or strategic importance, and regularly benchmarking performance against leading regional competitors rather than only global averages.

Innovation and Differentiation Opportunities

5. Sentiment-Driven Service Recovery

Using real-time sentiment monitoring powered by AI can help airlines identify dissatisfied passengers during their journey and allow frontline staff to address issues proactively before they escalate. Personalized recovery strategies based on passenger history and preferences will improve the effectiveness of service recovery. Immediate feedback mechanisms should be implemented to continuously identify and resolve small issues early.

6. Premium Experience Democratization

Extending elements of premium service into the economy cabin can differentiate airlines from budget competitors. This includes developing “economy plus” offerings that deliver select business-class features, implementing surprise service touches to delight economy passengers, training all crew to provide premium service regardless of cabin, and innovating cost-effective premium amenities suitable for economy passengers.

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